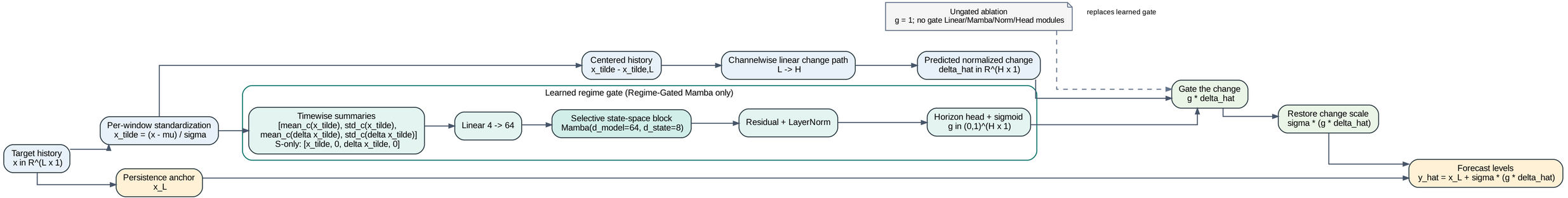
Target history
x in R^(L x 1)
Per-window standardization
x_tilde = (x - mu) / sigma
Persistence anchor
x_L
Centered history
x_tilde - x_tilde,L
Channelwise linear change path
L -> H
Predicted normalized change
delta_hat in R^(H x 1)
Learned regime gate (Regime-Gated Mamba only)
Timewise summaries
[mean_c(x_tilde), std_c(x_tilde),
mean_c(delta x_tilde), std_c(delta x_tilde)]
S-only: [x_tilde, 0, delta x_tilde, 0]
Linear 4 -> 64
Selective state-space block
Mamba(d_model=64, d_state=8)
Residual + LayerNorm
Horizon head + sigmoid
g in (0,1)^(H x 1)
Ungated ablation
g = 1; no gate Linear/Mamba/Norm/Head modules
replaces learned gate
Gate the change
g * delta_hat
Restore change scale
sigma * (g * delta_hat)
Forecast levels
y_hat = x_L + sigma * (g * delta_hat)